

Computational Oncology Study: p53 & KRAS G12D

 Python

 Biopython

 Jupyter

 License

 Status

Overview

This repository contains an independent computational oncology research project (May–June 2026) centered on a unified scientific theme: the **p53/KRAS G12D oncogenic axis** — two of the most critical proteins in human cancer biology.

The project spans sequence analysis, evolutionary conservation, multiple sequence alignment, and a comprehensive review of how bioinformatics unlocked the first clinical inhibitors of KRAS G12D — previously considered **undruggable for 40 years**.

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Background: Medical Biotechnology — Specialization toward Medical Data Science

Goal: PhD applications in Computational Oncology / Quantitative Biology (USA)

Scientific Context

Protein	Gene	UniProt	Role in Cancer
p53	TP53	P04637	Tumor suppressor – inactivated in ~50% of all cancers
KRAS (WT)	KRAS	P01116	Proto-oncogene – mutated in ~25% of all cancers
KRAS G12D	KRAS	–	Dominant mutation: ~42% PDAC, ~13% CRC, ~5% NSCLC

Repository Structure

```

KRAS-G12D-Computational-Oncology/
|
|— README.md
|
|— Task1_BLAST/
|   |— KRAS_Task1_BLAST.ipynb           # Python
notebook: sequence download + BLAST
|   |— KRAS_Task1_BLAST_Report.pdf      # 3-page
analysis report
|   |— p53_human.fasta                  # p53
sequence (UniProt P04637)
|   |— kras_human.fasta                  # KRAS WT
sequence (UniProt P01116)
|   |— blast_p53_results.xml             # Raw BLAST
XML output
|   |— blast_kras_results.xml            # Raw BLAST
XML output
|   |— blast_p53_identity.png            #
Conservation bar chart – p53
|   |— blast_kras_identity.png           #
Conservation bar chart – KRAS
|   |— blast_combined_results.csv        # Parsed
results table

```

```

|
├── Task2_MSA/
│   ├── KRAS_Task2_MSA.ipynb          # Python
│   └── KRAS_Task2_MSA_Report.pdf      # 3-page
alignment report
│   ├── kras_5species_input.fasta      # Multi-
FASTA input (5 species)
│   ├── kras_msa_aligned.fasta         # Clustal
Omega alignment output
│   ├── kras_msa_conservation.png      #
Conservation profile + G12 hotspot
│   └── kras_identity_matrix.png       # Pairwise
identity heatmap
|
└── Task3_DrugDiscovery/
    └── KRAS_Task3_DrugDiscovery_Report.pdf # 6-page
literature review report

```

Task 1 – DNA/Protein Sequence Analysis (BLAST)

Objective: Retrieve p53 and KRAS wild-type sequences from UniProt and perform comparative BLAST analysis against mammalian species to quantify evolutionary conservation.

Tools & Methods:

- UniProt REST API → programmatic FASTA retrieval
- NCBI blastp via `Bio.Blast.NCBIWWW.qblast()` (Biopython 1.85)
- Database: `refseq_protein` | Filter: `Mammalia[Organism]` | 15 hits per query
- Results parsed from XML → CSV + bar charts (Matplotlib)

Key Results:

Protein	Mean Identity	Min	Max	Hits at 100%
p53 (TP53)	98.0%	95.7%	100%	2/15
KRAS (WT)	99.7%	99.5%	100%	6/15

Key insight: KRAS is significantly more conserved than p53 across Mammalia. Glycine at position 12 (G12) – the G12D mutation hotspot – has been under extreme purifying selection for ~450 million years of vertebrate evolution, because any substitution impairs GTPase function.

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Task 2 – Multiple Sequence Alignment (MSA)

Objective: Align KRAS protein sequences from 5 vertebrate species using Clustal Omega, quantify per-position conservation, and demonstrate why G12 is an oncogenic hotspot.

Species selected:

#	Species	Common name	UniProt	Length	Role
1	<i>Homo sapiens</i>	Human	P01116	189 aa	Reference
2	<i>Mus musculus</i>	Mouse	P32883	189 aa	Cancer model
3	<i>Rattus norvegicus</i>	Rat	Q05144	192 aa	Oncology model

#	Species	Common name	UniProt	Length	Role
4	<i>Macaca mulatta</i>	Rhesus macaque	F7H3J8	290 aa	Primate outgroup
5	<i>Danio rerio</i>	Zebrafish	Q6NYN2	321 aa	Distant outgroup

Tools & Methods:

- Clustal Omega via EBI REST API
(<https://www.ebi.ac.uk/Tools/services/rest/clustalo>)
- Alignment: 429 positions | Job ID: `clustalo-R20260520-122300-0184-43298910-p2m`
- Per-position conservation scores computed via `Bio.AlignIO`
- G12D mutation computationally simulated on the human reference sequence

Key Results:

- Human–Mouse and Human–Rat identity: **98.9%**
- G12 (Glycine, position 12): **100% conserved** in canonical mammalian sequences
- G12D simulation: Glycine (57 Da, tiny) → Aspartic Acid (115 Da, charged) = GAP blocked → KRAS constitutively active

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Task 3 – Literature Review: Role of Bioinformatics in Drug Discovery

Objective: 6-page literature review demonstrating how bioinformatics made KRAS G12D – previously undruggable for 40

years — into a validated clinical target.

Topics covered:

- 1. The modern drug discovery pipeline and computational contributions
- 2. Sequence analysis & target identification (BLAST, MSA, UniProt)
- 3. Structural bioinformatics & AlphaFold (Nobel Prize 2024)
- 4. Virtual screening & molecular docking (AutoDock, GNINA)
- 5. Molecular dynamics simulations & free energy landscapes
- 6. Machine learning in drug discovery (DeepPurpose, GNINA CNN)

KRAS G12D Clinical Case Study:

Inhibitor	Mechanism	State	Status	Key Data
MRTX1133	Non-covalent, S-IIP	GDP-bound	Terminated Q1 2025 (BMS)	IC50 < 2 nM, 700× selectivity
Zoldonrasib (RMC-9805)	Covalent RAS-ON, CypA tri-complex	GTP-bound (active)	Phase I/1b active (NCT06040541)	30% ORR, 80% DCR in PDAC (2024)

Landmark result: Zoldonrasib achieved 30% Objective Response Rate and 80% Disease Control Rate in KRAS G12D-mutant pancreatic cancer patients (EORTC-NCI-AACR, Barcelona, October 2024) — a historic breakthrough for one of the deadliest cancers.

⚙️ Setup & Reproduction

Requirements

```
pip install biopython requests matplotlib numpy pandas
```

Run Task 1

```
# Open in JupyterLab or Jupyter Notebook
jupyter notebook Task1_BLAST/KRAS_Task1_BLAST.ipynb
```

⚠️ BLAST queries via NCBI web API take 2–5 minutes each. Do not interrupt the cells.

Run Task 2

```
jupyter notebook Task2_MSA/KRAS_Task2_MSA.ipynb
```

⚠️ Clustal Omega EBI submission takes ~60 seconds. Requires internet connection.

Environment tested on

- Python 3.9 | Biopython 1.85 | Anaconda JupyterHub (anaconda-2022.05-py39)
- NCBI blastp | EBI Clustal Omega REST API



Key References

1. Zeissig MN, et al. (2023). Targeting cancers with KRAS-G12D mutations. *Trends Cancer*, 9(11):955-67.
<https://pubmed.ncbi.nlm.nih.gov/37591766/>
2. Kondo Y, et al. (2023). Switch II pocket of KRAS(G12D) with monobody inhibitors. *PNAS*, 120(28):e2302485120.
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3. Wei D, et al. (2024). MRTX1133 targets KRAS G12D in PDAC. *Clin Cancer Res*, 30(4):655-662.
<https://pubmed.ncbi.nlm.nih.gov/37831007/>
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https://ascopubs.org/doi/10.1200/JCO.2025.43.4_suppl.724
 5. Jumper J, et al. (2021). AlphaFold. *Nature*, 596:583-589. [2024 Nobel Prize in Chemistry]
<https://www.nature.com/articles/s41586-021-03819-2>
 6. Sievers F, et al. (2011). Clustal Omega. *Mol Syst Biol*, 7:539.
<https://pubmed.ncbi.nlm.nih.gov/21988835/>
 7. UniProt Consortium. (2023). UniProt in 2023. *Nucleic Acids Res*, 51(D1):D523.
<https://academic.oup.com/nar/article/51/D1/D523/6835362>
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Independent research project, May–June 2026.